

Introduction To MDPs

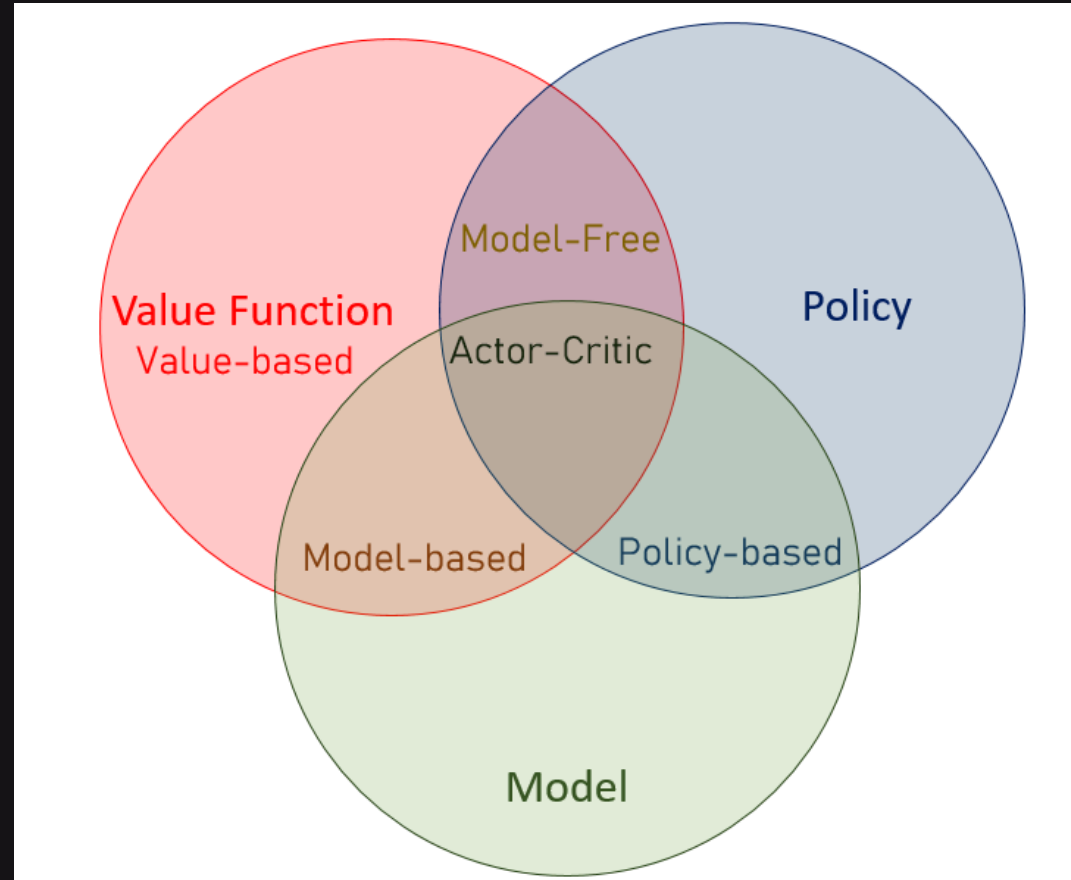
Instructor

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Classification of RL Agents



Markov Property

Imagine an aircraft at time t at position $x(t)$, traveling at const. speed v



$x(t)$



$$x(t + \delta) = x(t) + \delta v$$

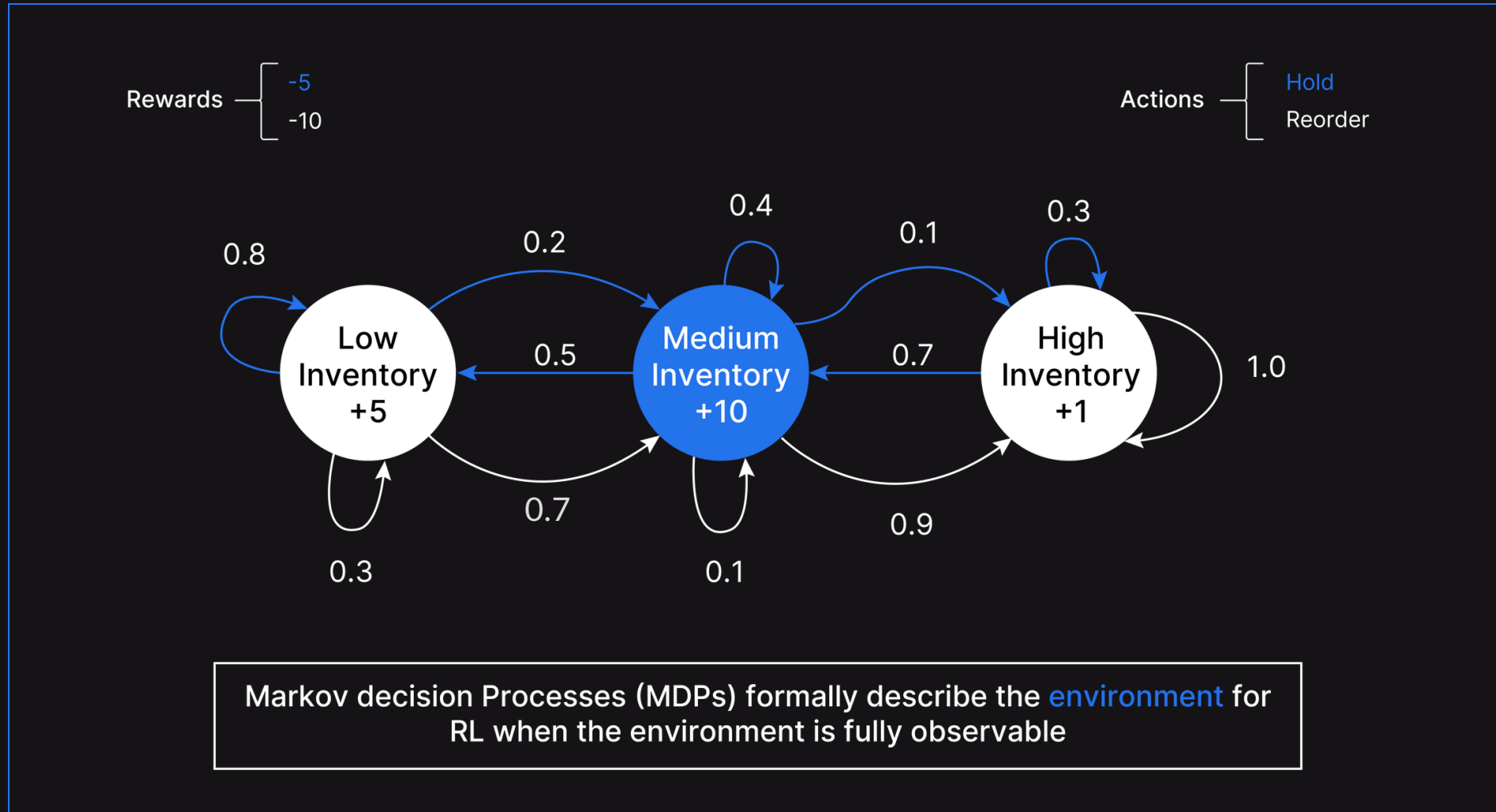
Markov Property: Its position at time $t + \delta$ is a function of ONLY its current position and not the past positions!

"The future is independent of the past given the present"

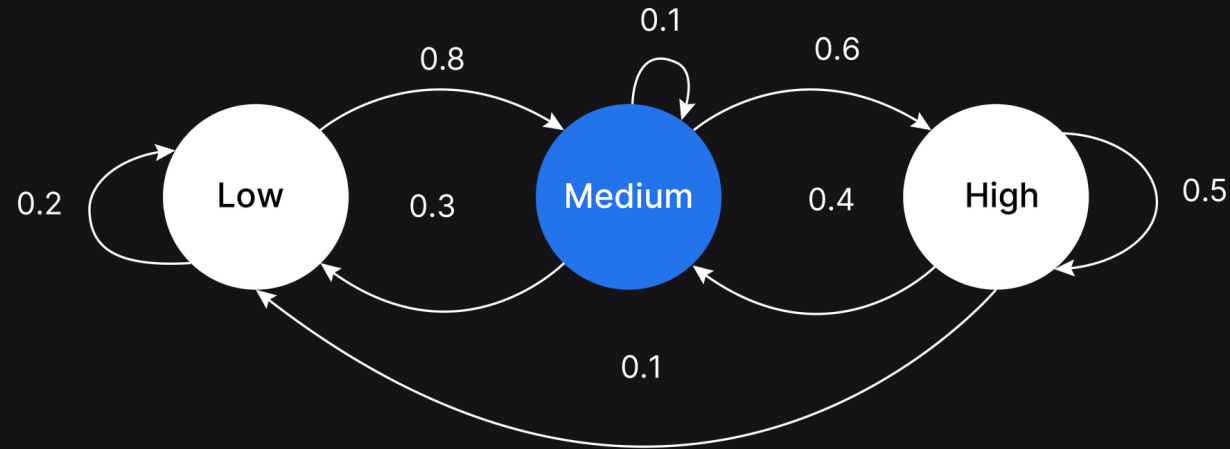
State S_t is Markov if and only if;

$$\mathbb{P}(S_{t+1} \mid S_t, S_{t-1}, \dots, S_1) = \mathbb{P}(S_{t+1} \mid S_t)$$

Markov Decision Process (MDP)



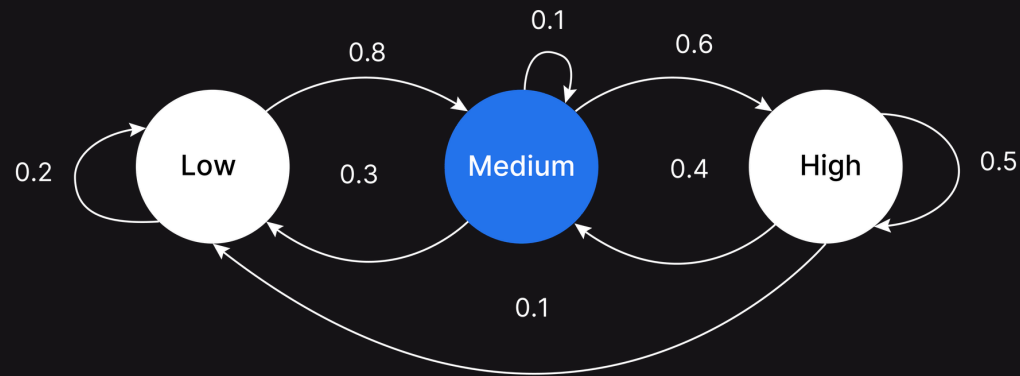
Markov Chain (Markov Process)



State transition probability from state s to s' is defined as:

$$\mathcal{P}_{ss'} = \mathbb{P}(S_{t+1} = s' \mid S_t = s)$$

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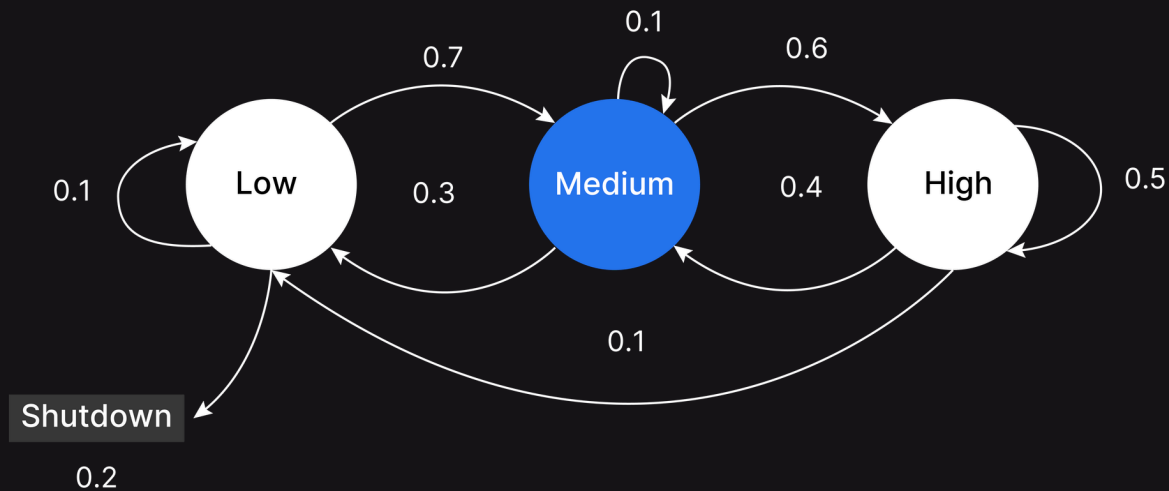
In this example, the state transition matrix is given by:

$$\mathcal{P} = \begin{bmatrix} \overset{\text{L}}{0.2} & \overset{\text{M}}{0.8} & \overset{\text{H}}{0} \\ \underset{\text{L}}{0.3} & \underset{\text{M}}{0.1} & \underset{\text{H}}{0.6} \\ \underset{\text{L}}{0.1} & \underset{\text{M}}{0.4} & \underset{\text{H}}{0.5} \end{bmatrix} \begin{matrix} \text{L} \\ \text{M} \\ \text{H} \end{matrix}$$

Where, the row sum is 1, and

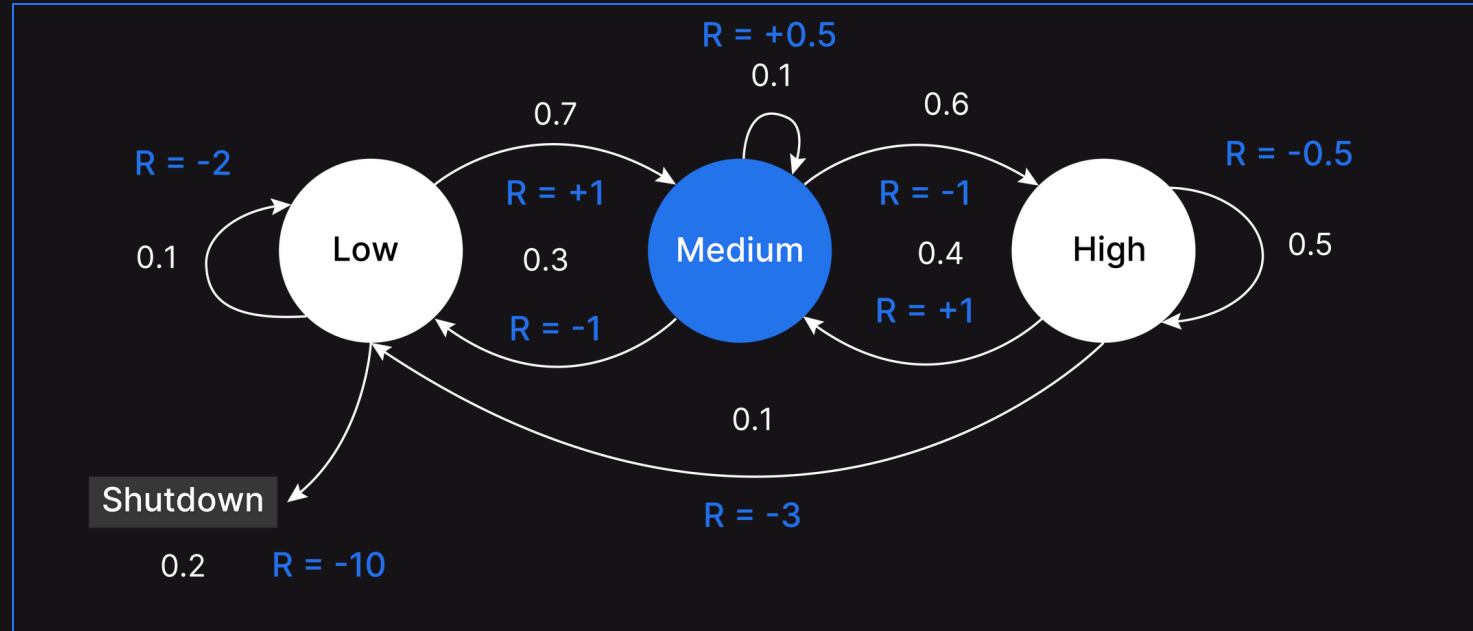
Markov Process is denoted by the tuple $\langle \mathcal{S}, \mathcal{P} \rangle$

Markov Chain (Markov Process) With Terminal State



- Shutdown is a terminal state.
- Reaching a terminal state is probable, making episodes finite.
- Sample episodes starting at medium:
 - Medium → High → High → Low → Medium → Low → Shutdown
 - Medium → Low → Low → Shutdown
 - Medium → High → Medium → High → Low → Medium → Low → Low → Shutdown

Markov Reward Process



- Transitions are associated with rewards
- Markov Reward Process (MRP) is denoted by $\langle \mathcal{S}, \mathcal{P}, \mathcal{R}, \gamma \rangle$
- Reward function; $\mathcal{R}$ is given by $\mathcal{R}_s = \mathbb{E}[R_{t+1} \mid S_t = s]$
- Discount factor, $\gamma \in [0, 1]$

Return

Return G_t is the total discounted reward from time-step t

$$G_t = R_{t+1} + \gamma R_{t+2} + \dots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

Myopic Agent

$$\gamma = 0$$

Far-sighted Agent

$$\gamma = 1$$

Why do we need discounting?

It is mathematically convenient, avoids potential infinite returns. Moreover, humans have preferences towards immediate reward over delayed reward.

Value Function (Expected Cost-To-Go)

The value function $v(s)$ of an MRP is the expected value of the return starting from state s

$$v(s) = \mathbb{E}[G_t \mid S_t = s]$$

$$G_1 = R_2 + \gamma R_3 + \gamma^2 R_4 + \cdots + \gamma^{T-2} R_T$$

Examples of sample episodes:

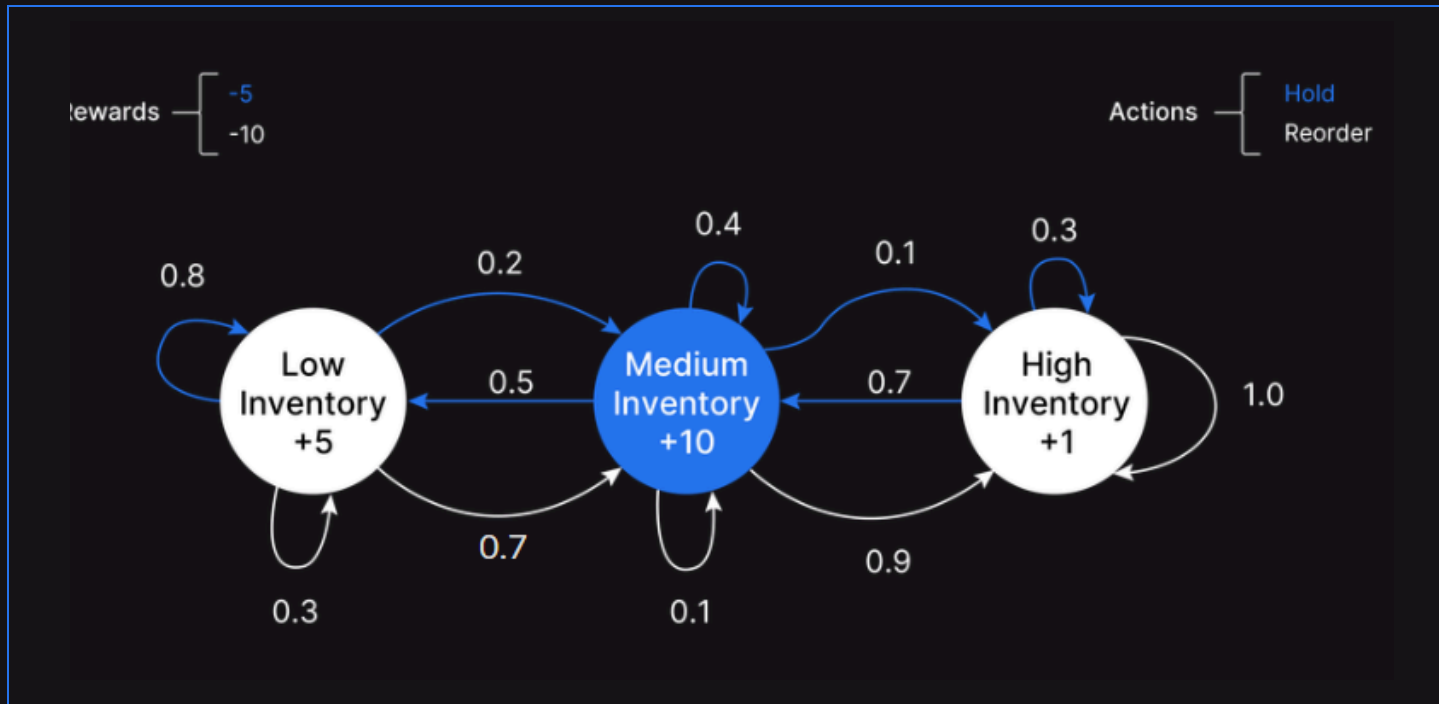
- M, H, H, L, M, L, shutdown
- M, L, L, shutdown
- M, H, L, L, shutdown

$$G_1 = -1 - \frac{1}{2} \cdot 0.5 - \frac{1}{4} \cdot 3 + \frac{1}{8} \cdot 1 - \frac{1}{16} \cdot 1 - \frac{1}{32} \cdot 10 = -2.25$$

$$G_1 = -1 - \frac{1}{2} \cdot 2 - \frac{1}{4} \cdot 10 = -4.5$$

$$G_1 = -1 - \frac{1}{2} \cdot 3 - \frac{1}{4} \cdot 2 - \frac{1}{8} \cdot 10 = -4.25$$

Markov Decision Process (Controlled MRP)



- Markov Decision Process (MRP) is denoted by $\langle S, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$

- Finite set of actions is represented by $\mathcal{A}$

- State transition probability is given by:

$$\mathcal{P}_{ss'}^a = \mathbb{P}[S_{t+1} = s' \mid S_t = s, A_t = a]$$

- Reward function is given by:

$$\mathcal{R}_s^a = \mathbb{E}[R_{t+1} \mid S_t = s, A_t = a]$$

Thank You
